AICA Level 2 — Capstone Project

AI-Assisted Year-End Adjustment Tracker with Automatic Income Tax Recomputation

Keeping final financial statements and the tax computation in agreement through every year-end revision

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Video: https://youtu.be/NVxZDCeZm5c | GitHub: aiinicai/AICA-Level-2-Projects (folder AICA-L2-Batch-101-TS-Surendranath)

Contents

1. Executive Summary

2. Problem Statement

3. Objectives

4. Tools Used

5. Solution Design

6. Use of Artificial Intelligence

7. Demonstration and Results

8. Impact: Time and Accuracy

9. Limitations

10. Future Scope

11. Conclusion

Annexure — Files Submitted

1. Executive Summary

At year end, the accounts of a company go through several rounds of changes: management's draft entries, the auditor's observations and the final closing entries. Every change to profit also changes the income tax computation, the current tax provision and deferred tax. When these are recomputed by hand in separate files, the final financial statements often do not agree with the final tax computation, and much time is spent redoing work.

This project builds a single Excel model, designed with the help of AI tools, in which every year-end adjustment is entered once in an Adjustment Register. The trial balance, Profit & Loss, Balance Sheet, income tax computation and deferred tax update automatically, and ten reconciliation checks confirm that everything agrees. AI (ChatGPT / Microsoft Copilot / Claude) is used to classify the tax treatment of each adjustment, map ledgers to financial statement lines, write the Excel formulas and draft client explanations.

On a demonstration company with 12 adjustments across three versions, profit before tax moved from ₹72.00 lakh to ₹47.75 lakh, total income was recomputed at ₹63.70 lakh and current tax at ₹16.03 lakh, with all ten checks passing. The illustrative time study shows recomputation time per revision falling from about 160 minutes to about 16 minutes.

2. Problem Statement

During finalisation of accounts, adjustments arrive late and in batches. Typical examples are:

- Depreciation corrections, prepaid and outstanding expenses, inventory write-downs to NRV
- Provisions for bonus, gratuity and doubtful debts
- Expenses where TDS was not deducted, penalties, interest to MSME suppliers
- Reclassifications between expense heads

Each adjustment can affect profit, the tax computation (through disallowances under sections 43B, 40(a)(ia), 36(1)(vii) and 37), current tax, deferred tax under AS 22, the provision for tax in the Balance Sheet and the net tax payable. In practice:

- Tax computations are revised manually in separate sheets, version after version.
- Profit in the computation is sometimes not updated to the final P&L figure.
- The provision for tax and deferred tax in the Balance Sheet may not match the final computation.
- There is no clear record of which change caused which movement in tax.

3. Objectives

1. Record every year-end adjustment in one register, with its version, accounting entry and tax nature.
2. Regenerate the adjusted trial balance, P&L and Balance Sheet automatically.
3. Recompute total income, current tax and deferred tax automatically from the final profit.
4. Provide automatic reconciliation checks between the financial statements and the tax computation.
5. Use AI tools to classify tax treatment, build formulas and explain changes.
6. Measure the time saved and the errors prevented.

Scope

The model covers a domestic company taxed under section 115BAA for FY 2025-26 (AY 2026-27), so section references follow the Income-tax Act, 1961. The Income-tax Act, 2025 applies from FY 2026-27 with new section numbers; the model's Tax Nature list can be relabelled for that. Data used is entirely dummy.

4. Tools Used

Tool	Purpose
Microsoft Excel	The working model: register, statements, computation, checks
ICAI AI Agent (multiple AI models)	Live classification of entries; comparing two models on the same prompts
Python (openpyxl)	Independent recomputation of profit and tax to cross-check the Excel model
Claude (Anthropic)	AI assistant used to help design the Excel model, write the Python script and draft this report; all output reviewed and finalised by the author
ChatGPT / Microsoft Copilot / Claude	Tax classification of entries, ledger mapping, formula writing, client notes
Excel features	SUMIFS, INDEX-MATCH, data validation drop-downs, conditional formatting
GitHub	Submission by fork and pull request to aiinicai/AICA-Level-2-Projects
Screen recorder (OBS / Loom / PowerPoint)	Video presentation with face camera

5. Solution Design

The workbook has eleven sheets that flow in one direction, so that an input is typed only once:

Sheet	What it does
Guide	How to use the model and the colour legend
Settings	Tax rates (22% + 10% surcharge + 4% cess = 25.168%) and the Tax Nature list
Trial_Balance	Unadjusted balances (input); adjustment Dr/Cr and adjusted balances (formulas)
Adjustments	One line per entry: version, Dr/Cr ledger, amount, Include Y/N, tax nature; impact on PBT, add-back, taxable income and tax are calculated
Financial_Statements	P&L (before, after, change) and Balance Sheet built from the adjusted trial balance
Tax_Computation	PBT → add-backs by type → less tax depreciation → total income → tax → deferred tax
Checks	Ten reconciliation checks with an overall ALL OK / REVIEW status, plus key figures
Version_Log	Movement in PBT, taxable income and tax for V1, V2 and V3
Time_Study	Manual vs. tool minutes per task
AI_Prompts	The prompts used, ready to copy
AI_Live_Demo	Four new entries and prompts for the live AI demonstration, with CA review points

5.1 The key idea: classify each entry by tax nature

Every adjustment is tagged with one of six tax natures. Each nature carries an add-back percentage and a difference type, which drive the tax computation and deferred tax automatically:

Tax nature	Add-back	Difference	Example
Allowable	0%	None	Prepaid insurance, NRV write-down
Disallowed - 43B (Timing)	100%	Timing (DTA)	Unpaid bonus, gratuity provision
Disallowed - 40(a)(ia) 30%	30%	Timing (DTA)	Fees without TDS
Disallowed - Provision (Timing)	100%	Timing (DTA)	Provision for doubtful debts
Disallowed - Permanent	100%	Permanent	Penalty, MSME interest
Book Depreciation	100%	Replaced	Depreciation correction

5.2 How the impact of an entry is calculated

For each entry, the model looks up whether the debit and credit ledgers are P&L items. Impact on PBT = amount if the credit side is a P&L ledger, minus amount if the debit side is a P&L ledger. For an expense, the tax add-back = impact × add-back %. Impact on taxable income = PBT impact + add-back, and impact on tax = taxable impact × effective rate.

6. Use of Artificial Intelligence

AI was used as an assistant at five stages. Every AI output was reviewed and verified by me before use.

Stage	What AI did	Result in the model
Tax classification	Classified each journal entry into one of the six tax natures with section reference and DTA/DTL effect	Tax Nature and AI suggestion columns
Ledger mapping	Mapped ledger names to Schedule III lines and P&L / BS types	Type and FS Line columns
Formula building	Wrote SUMIFS, INDEX-MATCH and the impact formula compatible with Excel 2016	All calculation columns
Check design	Listed reconciliation checks between accounts and tax computation	Checks sheet
Communication	Drafted plain-language notes explaining changes between versions	Client note shown in the demo

Sample prompt — tax classification

> *You are a Chartered Accountant in India. For FY 2025-26 (Income-tax Act, 1961), classify each year-end journal entry below into exactly one of: Allowable; Disallowed - 43B (Timing); Disallowed - 40(a)(ia) 30%; Disallowed - Provision (Timing); Disallowed - Permanent; Book Depreciation. Give the section reference and whether it creates a DTA, DTL or permanent difference. Return a table.*

All six prompts are in the AI_Prompts sheet of the workbook.

Human review of AI output

AI suggestions were checked against the Act before being accepted. For example, late-payment charges need care: a penalty for violating a law is disallowed under Explanation 1 to section 37(1), while interest on delayed GST is compensatory and generally allowable, so such items are flagged for separate verification. This review step is part of the method, not an afterthought.

6.1 Live AI demonstration (shown in the video)

To show AI as a working part of the process rather than a background aid, four new last-minute entries are classified live by an AI assistant on camera, then reviewed and entered into the model. Two of them were chosen deliberately because they test whether the AI links different provisions correctly:

New entry	Correct treatment	CA review point
Leave encashment provision ₹1,50,000	43B (Timing)	Allowed on payment u/s 43B(f); creates DTA
Interest on delayed GST ₹18,000	Allowable	Compensatory interest, not a penalty; an AI may wrongly treat it as penal
Agent commission without TDS ₹2,00,000	40(a)(ia) — 30%	₹60,000 disallowed; allowed when TDS is paid
Donation to political party ₹50,000	Permanent disallowance	An AI may suggest 80GGB, but a 115BAA company cannot claim Chapter VI-A deductions (other than 80JJAA / 80M)

The four entries are kept ready in the register, switched off. When their Include flag is changed from N to Y, the model recomputes instantly: PBT falls from ₹47.75 lakh to ₹43.57 lakh, total income from ₹63.70 lakh to ₹62.12 lakh, current tax from ₹16.03 lakh to ₹15.63 lakh, and the deferred tax credit rises to ₹3.78 lakh. All ten checks remain OK.

AI is then given the Version Log and asked to write a short note to the client's managing director explaining why profit and tax changed. This shows AI in a second role: communication.

6.2 Human-in-the-loop principle

The project follows a clear division of work: AI produces a fast first draft (classification, formulas, explanations); the Chartered Accountant reviews, corrects where needed and takes the final decision; Excel performs the calculations and the checks. This keeps the speed of AI while retaining professional accountability.

The actual answers given by the AI models in the live test are recorded in section 6.3.

6.3 Second judgement test and two-model comparison

A second test checks whether AI links section 115BAA with additional depreciation. New looms costing ₹40,00,000 were put to use in February 2026, so only half the normal rate applies. A model that misses the link would claim 20% additional depreciation u/s 32(1)(iia), which a 115BAA company cannot claim.

Item	Likely AI answer (₹)	Correct (₹)
Normal depreciation 15% × ½	3,00,000	3,00,000
Additional depreciation 20% × ½	4,00,000	Not allowed
Total depreciation	7,00,000	3,00,000

Accepting the wrong answer would understate tax by about ₹1,00,672 (₹4,00,000 × 25.168%). Both judgement tests are run on two different models through the ICAI AI Agent, and their answers are compared on camera.

Test	Model 1: Gemini 3.8 Flash (ICAI AI Agent)	Model 2: GPT-6.1 Sol (ICAI AI Agent)
Political donation ₹50,000 (115BAA vs 80GGB)	Correct: fully disallowed; noted 80GGB barred under 115BAA	Correct: fully disallowed; 80GGB unavailable under 115BAA(2)(i)
Additional depreciation on looms (115BAA)	Correct: ₹3,00,000 only; additional depreciation excluded	Correct: ₹3,00,000 only; cited 115BAA(2)(i) and (iv)
Interest on delayed GST ₹18,000	Correct: allowable as compensatory u/s 37(1)	Correct: allowable as compensatory u/s 37(1)
CA's final decision (entered in the model)	Disallowed; ₹3,00,000; allowable	Disallowed; ₹3,00,000; allowable

Both models reached the correct conclusion on all three tests. Their approach differed: Gemini 3.8 Flash supported the GST interest point with Supreme Court case law, while GPT-6.1 Sol cited the Income-tax Department's website, stated its assumptions, and added qualifications such as the payee-compliance relief under section 201(1) and the recognition criteria for a deferred tax asset. Before the conclusions were entered in the model, each one, including the section and case citations, was verified against the Act.

The test shows that current AI models handle these points well. Professional review still matters, because an answer can change with the wording of the prompt or the model used, and responsibility for the tax position stays with the Chartered Accountant.

6.4 Independent check with Python

A short Python script (check_model.py) reads the trial balance and adjustment register from the workbook and recomputes profit before tax, total income, current tax and deferred tax without using any Excel formula. It then compares its answers with the Excel model. On the demonstration data all four figures match, and on the data with the four live-demo entries added they match again (PBT ₹43,57,000; tax ₹15,63,440).

6.5 Responsible use of AI

As required by ICAI, no confidential, personal or client information was entered into any AI platform. All data in this project is dummy data created for demonstration. Every AI suggestion was reviewed by the author before use, and AI-generated tax views are treated as drafts, not as conclusions.

6.6 Course learning applied

The project applies the following learnings from AICA Level 2, Batch L2B101, Chennai (online 10–11 October 2026; physical 12–14 October 2026):

Course learning	How it was used in this project
ICAI AI Agent	Used to run the same tax-classification prompts on two different AI models and compare their answers live.
Prompt engineering	Wrote structured prompts with a role, fixed answer categories and a table format, so that AI answers drop straight into the model.

Course learning	How it was used in this project
Python	Used check_model.py (prepared with AI assistance), which recomputes profit and tax independently and confirms the Excel figures.
Excel / Google Sheets	Built the linked model with SUMIFS, INDEX-MATCH, drop-down lists and automatic checks; it also opens in Google Sheets.
Responsible AI	Used only dummy data and treated every AI answer as a draft for professional review.

7. Demonstration and Results

The demonstration uses a dummy company, Sunrise Textiles Pvt Ltd, with 26 ledgers and 12 year-end adjustments in three versions.

7.1 Adjustments entered

No.	Ver.	Adjustment	Amount (₹)	Tax nature
A01	V1	Depreciation on new looms not charged	3,50,000	Book Depreciation
A02	V1	Prepaid insurance	1,20,000	Allowable
A03	V1	Audit fee provision	1,50,000	Allowable
A04	V2	Bonus payable in Nov 2026	4,00,000	43B (Timing)
A05	V2	Gratuity provision	6,00,000	43B (Timing)
A06	V2	Consultancy fee — no TDS	3,00,000	40(a)(ia) 30%
A07	V2	Inventory write-down to NRV	4,50,000	Allowable
A08	V2	Provision for doubtful debts	2,50,000	Provision (Timing)
A09	V3	Interest to MSME supplier	80,000	Permanent
A10	V3	Penalty for late filing	25,000	Permanent
A11	V3	Accrued FD interest	60,000	Allowable
A12	V3	Rent reclassification	1,00,000	Allowable

7.2 Movement through versions

Version	Stage	PBT impact (₹)	Cumulative PBT (₹)	Tax impact (₹)
V0	Unadjusted	-	72,00,000	-
V1	Draft entries	(3,80,000)	68,20,000	(7,550)
V2	Audit observations	(20,00,000)	48,20,000	(1,66,109)
V3	Final close	(45,000)	47,75,000	15,101

Although audit entries reduced profit by ₹20 lakh in V2, current tax fell by only about ₹1.66 lakh, because most of those entries were disallowed (43B, 40(a)(ia), provisions). This is exactly the kind of insight that is hard to see in manual worksheets.

7.3 Final tax computation

Particulars	Amount (₹)
Profit before tax as per P&L	47,75,000
Add: Depreciation as per books	18,50,000

Particulars	Amount (₹)
Add: CSR expenditure	2,00,000
Add: Disallowed u/s 43B	10,00,000
Add: Disallowed u/s 40(a)(ia) (30%)	90,000
Add: Provision for doubtful debts	2,50,000
Add: Permanent disallowances	1,05,000
Less: Depreciation as per Income-tax Act	(19,00,000)
Total income	63,70,000
Current tax @ 25.168%	16,03,200
Less: Advance tax & TDS	(14,00,000)
Net tax payable	2,03,200
Deferred tax credit (net timing difference ₹12,90,000 × 25.168%)	(3,24,667)

7.4 Reconciliation checks — all OK

1. Unadjusted trial balance totals to zero
2. Adjusted trial balance totals to zero
3. Every adjustment uses a valid ledger code
4. Every adjustment has a tax nature
5. Balance Sheet tallies (₹3,84,05,000 both sides)
6. PBT in tax computation equals PBT in P&L
7. Current tax in P&L equals the computation
8. Provision for tax in Balance Sheet equals the computation
9. Deferred tax movement in Balance Sheet equals the P&L charge
10. Sum of adjustment impacts equals the change in PBT

7.5 Live what-if test

Changing Include for A05 (gratuity, ₹6 lakh) from Y to N raises PBT by ₹6 lakh, leaves current tax unchanged (because it was a 43B add-back) and reduces the deferred tax credit — all instantly, with all checks still OK. This is also shown in the video.

8. Impact: Time and Accuracy

Task (per revision)	Manual (min)	Tool (min)	Saved
Pass entries and update trial balance	20	5	75%
Redraft P&L and Balance Sheet	30	0	100%
Identify tax treatment of new entries	25	5	80%
Recompute total income and tax	30	0	100%
Recompute deferred tax	20	0	100%
Tie tax provision & DTL to statements	15	1	93%
Explain changes to partner / client	20	5	75%
Total per revision	160	16	90%

With three revisions per close, this is about 7.2 hours saved per client per year. [Replace these illustrative timings with your own measured figures before submission.]

Accuracy benefits

- Single source of truth: each entry is typed once, so the statements and computation cannot drift apart.
- Ten automatic checks catch mismatches before signing.
- Every movement in tax is traceable to a specific adjustment and version.
- The AI suggestion column documents the reasoning for each tax treatment, which helps review and audit.

9. Limitations

- AI tax suggestions can be wrong or outdated; professional verification is essential.
- The model covers common adjustments; items like MAT, carry-forward losses, deductions under Chapter VI-A and interest u/s 234B/234C are not yet included.
- Tax depreciation is a single input; a full block-wise schedule is not built in.
- Data must be exported from the accounting software into Excel.

10. Future Scope

- Direct import of trial balance from Tally Prime or ERP.
- Interest u/s 234B/234C and MAT / normal-provisions comparison.
- Block-wise tax depreciation schedule.
- Relabelling of the Tax Nature list for the Income-tax Act, 2025 from FY 2026-27.
- Copilot in Excel to suggest tax nature automatically as each entry is typed.

11. Conclusion

The project shows how a Chartered Accountant can combine AI assistance with a well-structured Excel model to solve a real, recurring year-end problem. By recording each adjustment once and letting the statements, tax computation and deferred tax flow from it, the final accounts and the tax computation stay in agreement through every revision, with large time savings and a clear audit trail. AI speeds up classification, formula building and communication, while professional judgement remains with the CA.

Annexure — Files Submitted

File	Description
YearEnd_Tax_Recompute_Model.xlsx	Working Excel model (11 sheets, including AI_Live_Demo)
Project_Report.docx / .pdf	This report
Video_Script.md	Script for the video presentation
README.md	Project overview on GitHub
check_model.py	Python script that independently recomputes profit and tax and compares with Excel